PROJECT TITLE

AI-Based Health Insurance Premium Prediction & Risk Analysis using Explainable AI

INTRODUCTION

This project predicts the annual medical insurance premium of a person using machine learning models. Insurance cost depends on personal and lifestyle factors such as age, BMI, smoking habits, region, and number of children. The system provides not only premium prediction but also a risk category and explainable AI insights to improve transparency in healthcare insurance decisions.

PROBLEM STATEMENT

Traditional insurance premium calculation is complex and varies based on risk factors. Manual estimation is inaccurate and time-consuming. This project aims to automate the premium prediction process using machine learning and provide clear explanations for why the predicted premium is high or low.

OBJECTIVES

- Predict medical insurance charges using ML regression models
- Classify users into low/medium/high risk categories
- Provide explainable outputs using feature importance
- Store prediction history for future tracking
- Generate downloadable insurance prediction reports

SYSTEM ARCHITECTURE (Same Review Pattern)

Workflow:

User Input → Data Preprocessing → ML Model Prediction → Risk Classification → Report Generation → History Storage

Example:

1. User enters age, BMI, smoker status
2. Model predicts premium
3. Risk level displayed
4. Prediction saved in history
5. PDF report generated

DATASET DESCRIPTION

Dataset used: **Medical Insurance Dataset**

Attributes:

- age
- sex
- bmi
- children
- smoker
- region
- charges (target variable)

This dataset helps train the regression model for predicting insurance cost.

MACHINE LEARNING MODEL USED

Random Forest Regression

Random Forest is chosen because:

- Handles non-linear relationships
- Provides high accuracy
- Works well with mixed features

Model is saved as:

- rf_model.pkl
- scaler.pkl

DATA PREPROCESSING

To make data model-ready:

- Categorical values encoded
    - male → 1, female → 0
    - yes → 1, no → 0
- Numerical features scaled using StandardScaler

This improves prediction quality.

RISK ANALYSIS MODULE (Final Year Upgrade)

Based on predicted premium:

- Low Risk → Basic Plan
- Medium Risk → Standard Plan
- High Risk → Premium Plan

This makes the project a risk management system, not just prediction.

EXPLAINABLE AI COMPONENT

The model outputs feature importance showing which factor impacts premium most:

- Smoking increases premium strongly
- Higher BMI increases risk
- Age contributes significantly

This provides transparency similar to Explainable AI methods mentioned in Review–II.

IMPLEMENTATION PLATFORM

Technology Stack:

- Python
- Flask Web Framework (no Streamlit)
- HTML Templates
- Scikit-learn
- Pickle
- PDF Report Generator (FPDF)
- CSV-based Prediction History

MODULES IN PROJECT

Module 1: User Input Web Portal

User enters personal details.

Module 2: Premium Prediction

ML model predicts annual charges.

Module 3: Risk Classification

Low / Medium / High insurance risk.

Module 4: Prediction History

All predictions stored in prediction_history.csv

Module 5: PDF Report Generation

User report generated automatically.

TESTING & EVALUATION

Evaluation metrics used:

- Accuracy (R^2 score)
- Mean Absolute Error
- Model validation using train-test split

Testing includes:

- Unit testing (input validation)
- Integration testing (model + UI)
- System testing (full prediction workflow)

RESULTS

Output Screenshots:

Prediction Result Page
Risk Category Output
History Page
Generated PDF Report